Solution:

Step 1 : Let μ and σ represent respectively the mean and standard deviation of the probability distribution of the life span of the bulbs. It is given that σ = 150 hours. The null and alternative hypotheses are

Null hypothesis: H_0: μ = 2000

i.e., the mean life span of the bulbs is not significantly different from 2000 hours.

Alternative hypothesis: H_1 : $\mu \neq 2000$

i.e., the mean life span of the bulbs is significantly different from 2000 hours.

It is a two-sided alternative hypothesis.

Step 2 : **Data**

The given sample information are

Sample size (n) = 100, Sample mean ($\bar{x}$) = 1950 hours

Step 3 : **Level of significance**

α = 5%

Step 4 : **Test statistic**

The test statistic is $Z = \dfrac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$, under H_0

Under the null hypothesis H_0, Z follows the $N(0,1)$ distribution.

Step 5 : **Calculation of Test Statistic**

The value of Z under H_0 is calculated from

$$z_0 = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

as

$$z_0 = \frac{1950 - 2000}{150 / \sqrt{100}}$$

$$= -3.33$$

Thus; $|z_0| = 3.33$

Step 6 : **Critical value**

Since H_1 is a two-sided alternative, the critical value at α = 0.05 is $z_e = z_{0.025} = 1.96$. (*see* Table 1.6).

Step 7 : **Decision**

Since H_1 is a two-sided alternative, elements of the critical region are determined by the rejection rule $|z_0| \geq z_e$. Thus, it is a two-tailed test. For the given sample information, the rejection rule holds *i.e.*, $|z_0| = 3.33 > z_e = 1.96$. Hence, H_0 is rejected in favour of H_1: $\mu \neq 2000$. Thus, the mean life span of the LED bulbs is significantly different from 2000 hours.

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